# Comprehensive Unload.dll Malware Analysis

## Analyst Information

- **Analyst:** Ahmed Emad Nasr
- **Date:** April 5, 2026
- **Target Artifact:** unload.dll
- **Suspected Threat Family:** IcedID (BokBot)

Ahmed Emad Nasr

---

## 1. Executive Summary

This report presents a **static analysis** of the DLL file unload.dll, which exhibits multiple indicators consistent with **IcedID malware behavior**.

Key findings confirm that the binary:

- Is **obfuscated and partially packed**
- Uses **WinHTTP APIs** for outbound communication
- Implements **dynamic API resolution** for evasion
- Contains a **hardcoded C2 domain**
- Likely acts as a **loader for second-stage payloads**

---

## 2. File Metadata & Identification

| Attribute | Value |
|---|---|
| File Name | unload.dll |
| Original Name | loader_dll_64.dll |
| File Size | 15.94 KiB (16,320 bytes) |
| SHA-256 | A3FA68045D0106D6DB3D43DF6B5997D9034F9F7D2A34148187498E4B504EBF58 |
| Architecture | 64-bit (AMD64) |
| File Type | PE64 DLL |
| Subsystem | Windows GUI |
| Compiler | MSVC 19.12 |
| Linker | 14.12 |
| Toolset | Visual Studio 2017 |

| Attribute | Value |
|---|---|
| Timestamp | 2022-11-30 20:23:22 UTC |
| Entropy | 4.665 |
| Overlay | Present (Offset: 0x3400, Size: 0x0bc0) |

**Analysis Notes**

- **Moderate entropy (4.665)** indicates possible **partial packing or obfuscation**
- **Overlay presence** strongly suggests **appended encrypted payload/configuration**
- Compilation metadata aligns with **legitimate toolchains**, commonly abused by malware authors

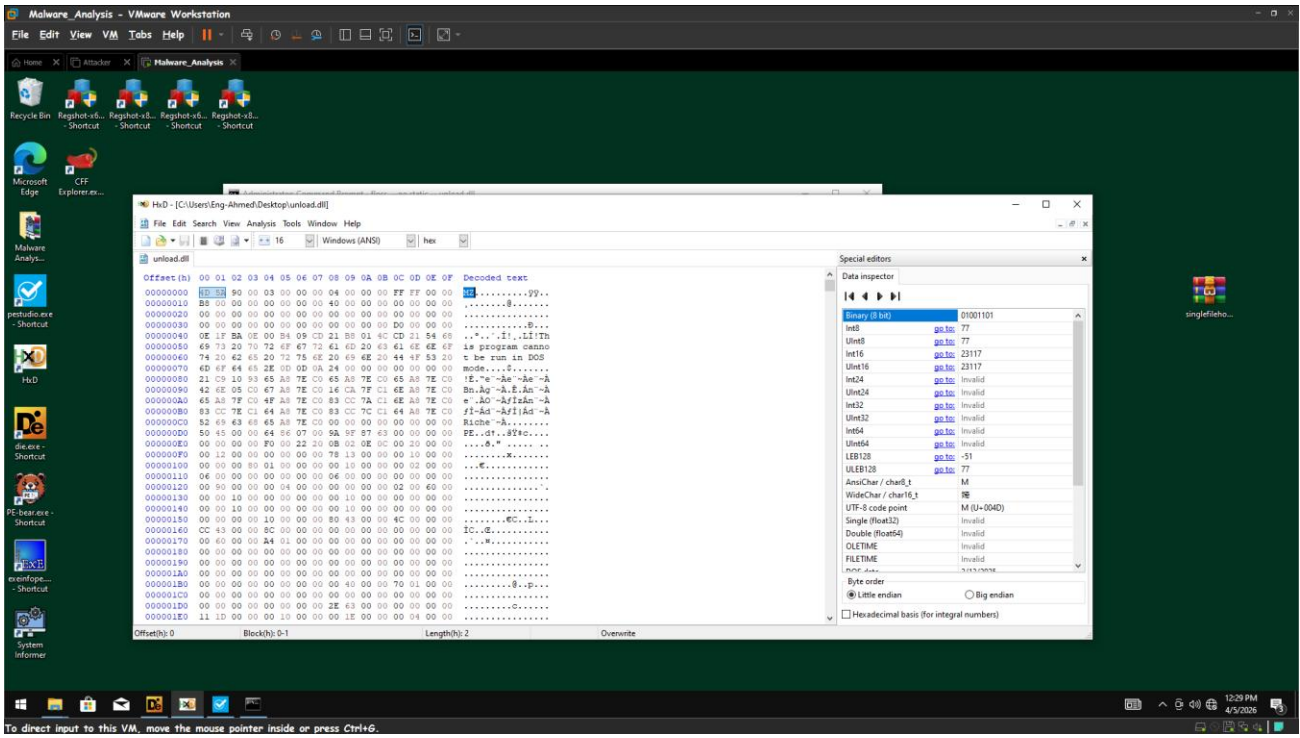
---

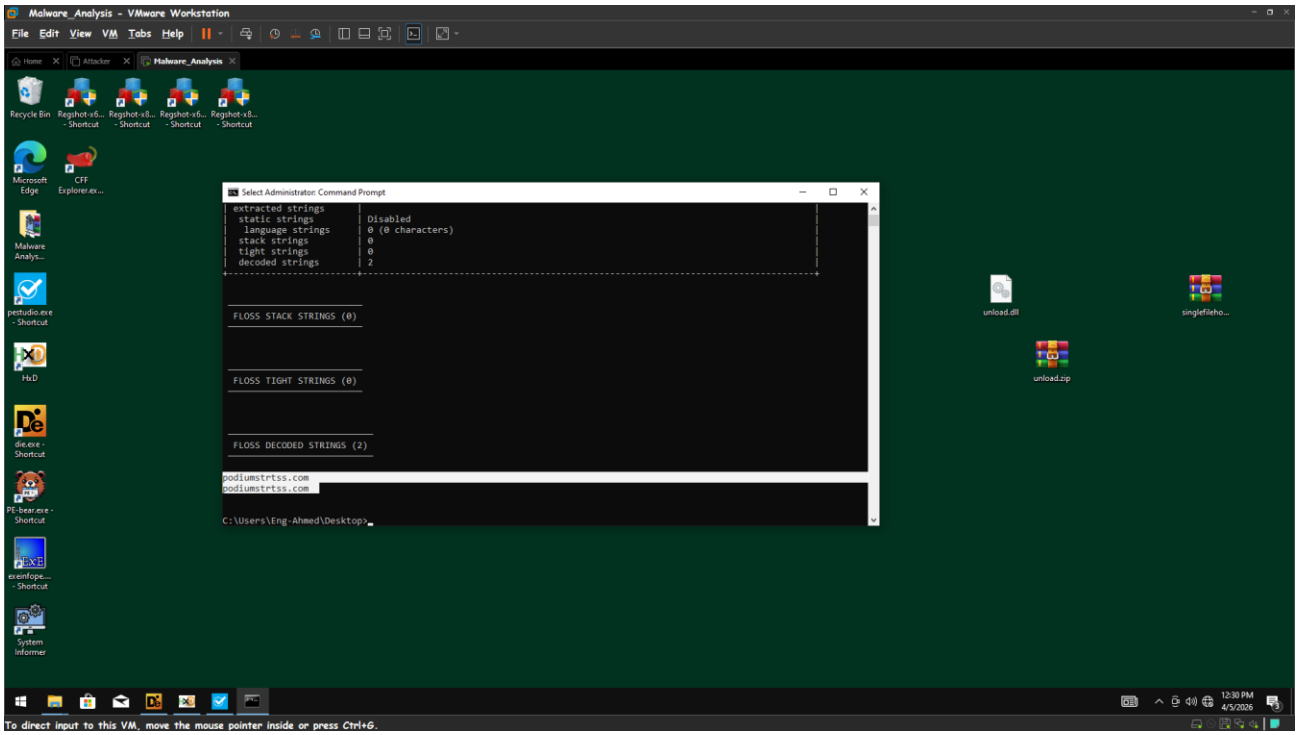
**3. PE Section Analysis**

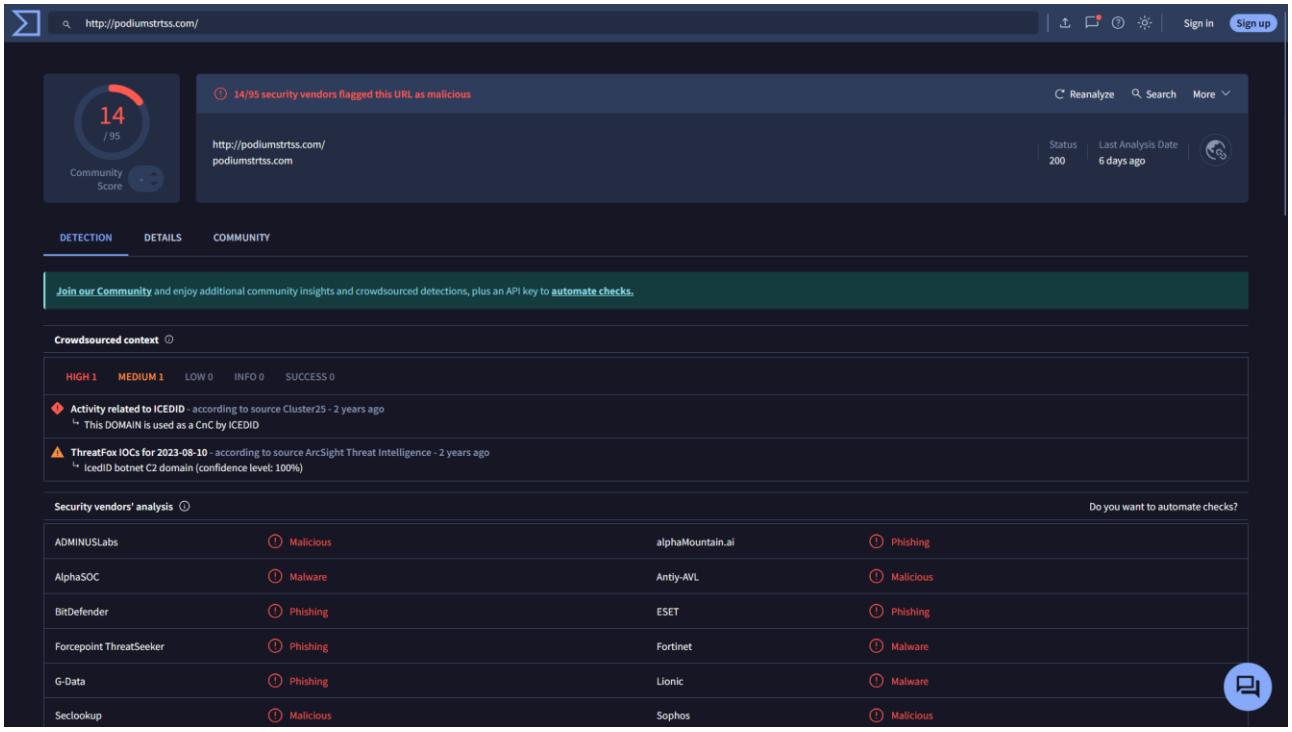
| Section | Entropy | Characteristics | Notes |
|---|---|---|---|
| .c | 6.153 | Executable, Read | Highly suspicious (likely obfuscated code) |
| .text | 0.082 | Executable, Read | Extremely small → likely stub |
| .rdata | 4.143 | Read-only | Contains imports/strings |
| .data | N/A | RW | Virtual section |
| .pdata | 3.415 | Read | Exception handling |
| .r | 3.399 | Read | Non-standard |
| .d | 2.443 | RW | Non-standard |

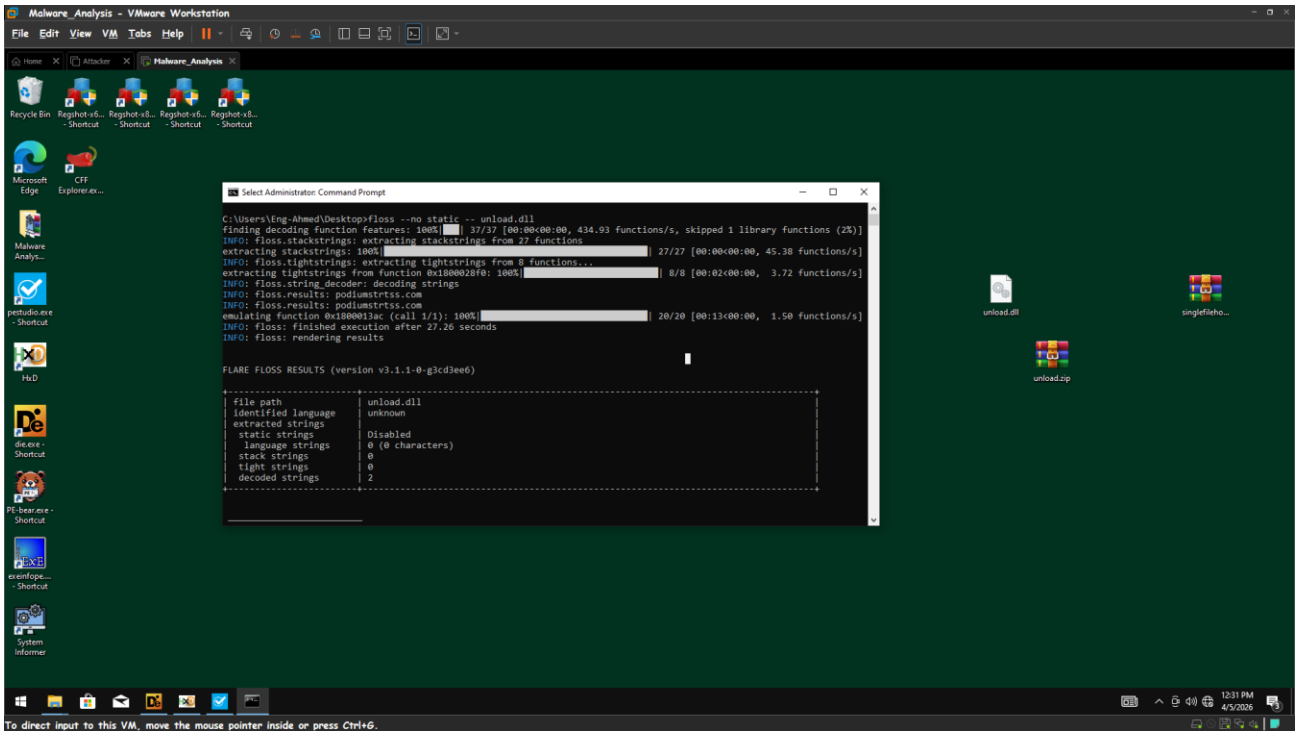
**Analysis Notes**

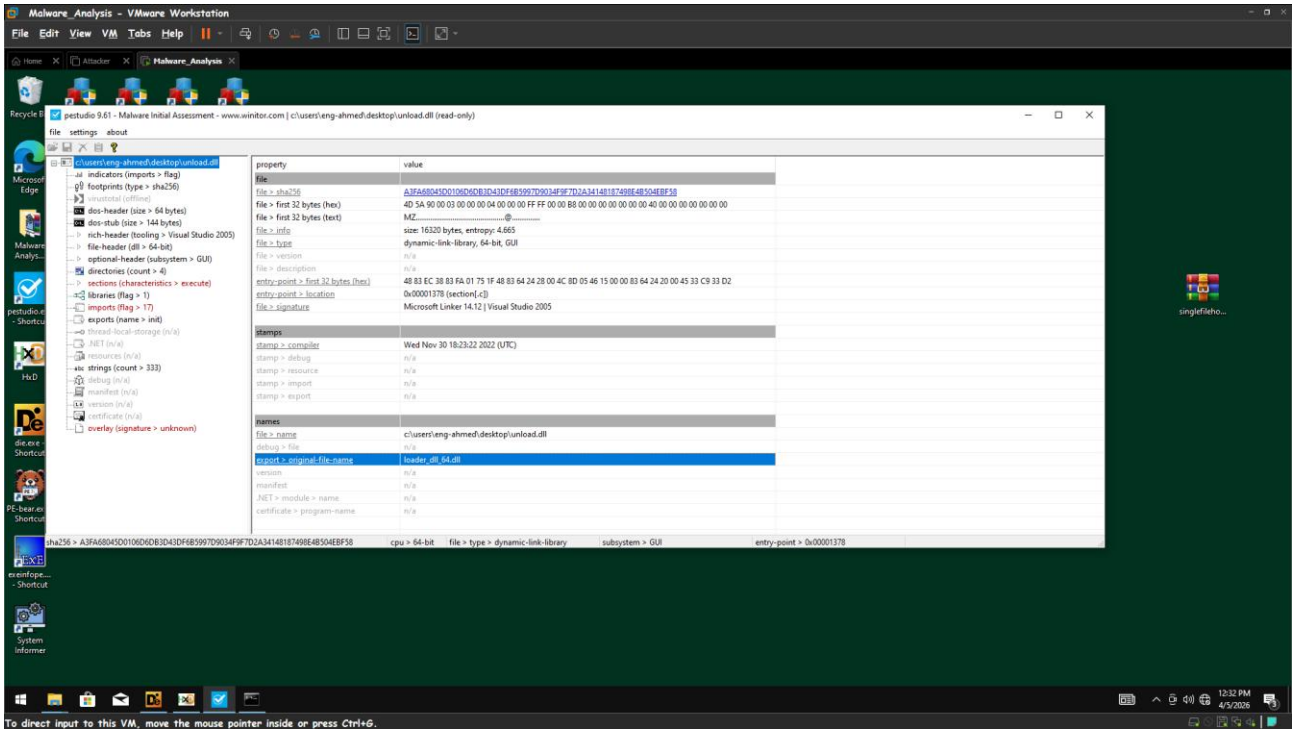
- Use of **non-standard section names** is a strong evasion indicator
- **.c section replacing .text** suggests **manual manipulation or packing**
- **Tiny .text section** indicates **execution redirection to hidden logic**

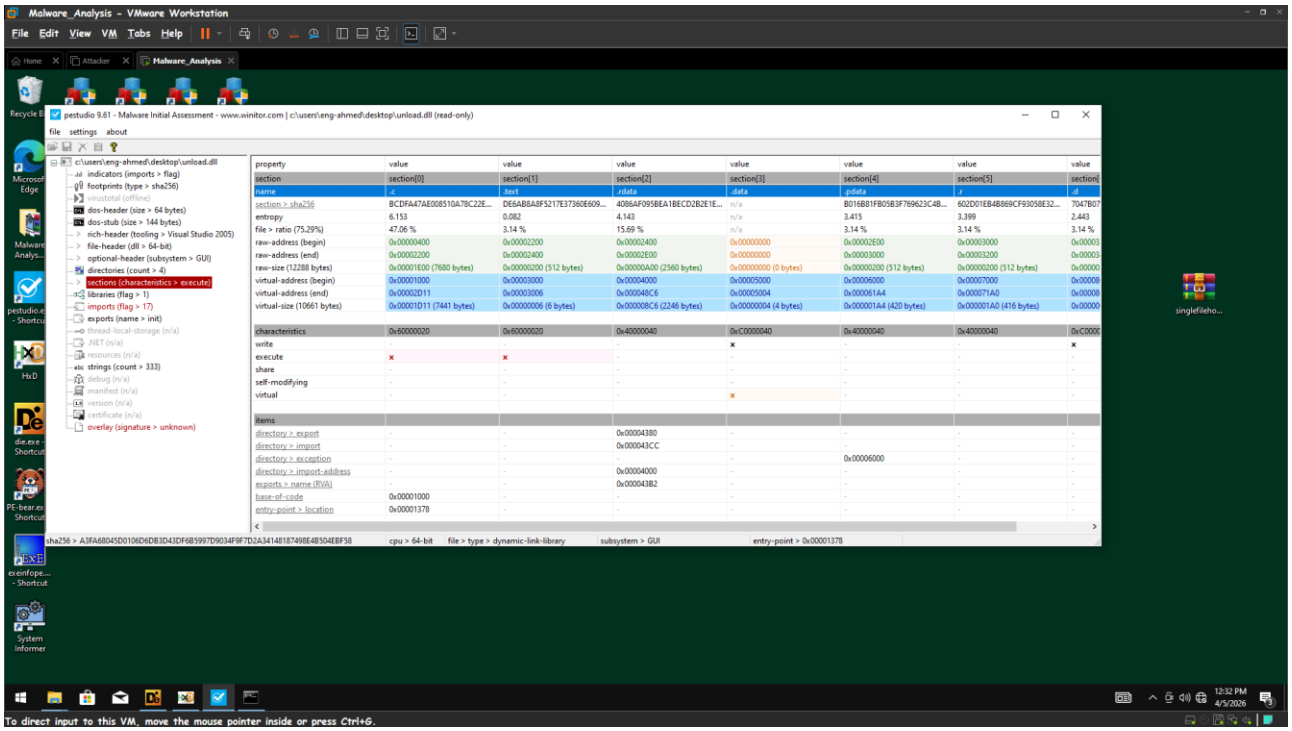

Malware_Analysis - VMware Workstation
File Edit View VM Tabs Help
HxD - [C:\Users\Eng-Ahmed\Desktop\unload.dll]
File Edit Search View Analysis Tools Window Help
unload.dll
Offset(h) 00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F Decoded text
Special editors
Data inspector
Byte order
Little endian
Big endian
Hexadecimal basis (for integral numbers)
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

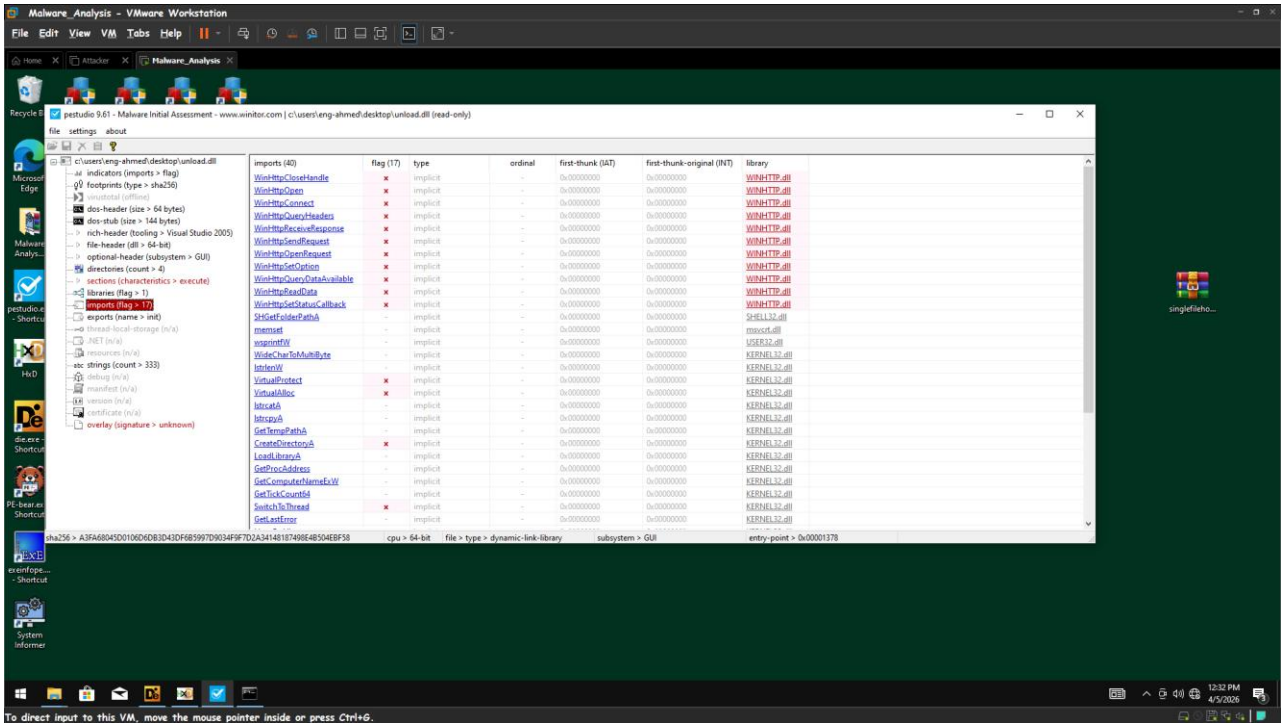

Malware_Analysis - VMware Workstation
File Edit View VM Tabs Help
Select Administrator: Command Prompt
extracted strings
static strings
Disabled
language strings
0 (0 characters)
stack strings
tight strings
decoded strings
FLOSS STACK STRINGS (0)
FLOSS TIGHT STRINGS (0)
FLOSS DECODED STRINGS (2)
podiumstrtss.com
podiumstrtss.com
C:\Users\Eng-Ahmed\Desktop>
unload.dll
unload.zip
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

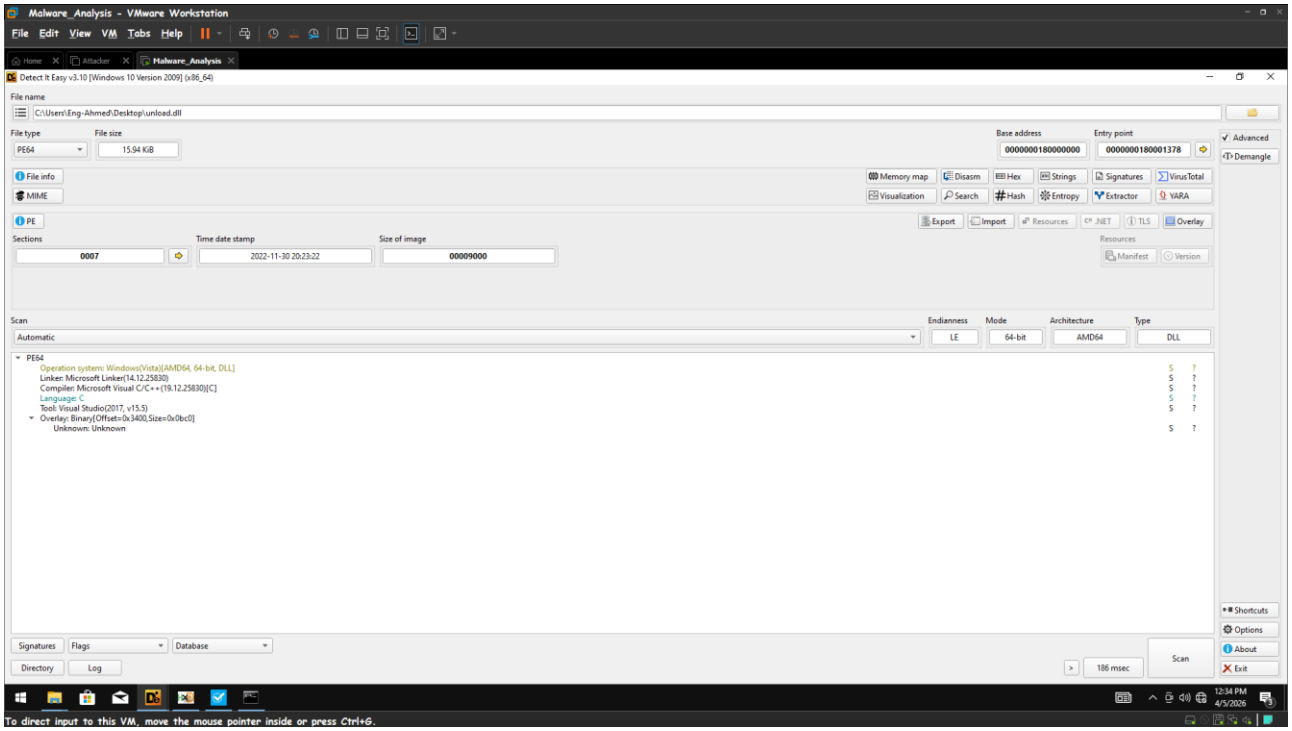

http://podiumstrtss.com/
Sign in
Sign up
14
/ 95
Community Score
14/95 security vendors flagged this URL as malicious
Reanalyze
Search
More
http://podiumstrtss.com/
podiumstrtss.com
Status
200
Last Analysis Date
6 days ago
DETECTION
DETAILS
COMMUNITY
Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.
Crowdsourced context
HIGH 1
MEDIUM 1
LOW 0
INFO 0
SUCCESS 0
Activity related to ICEDID - according to source Cluster25 - 2 years ago
This DOMAIN is used as a CnC by ICEDID
ThreatFox IOCs for 2023-08-10 - according to source ArcSight Threat Intelligence - 2 years ago
IcedID botnet C2 domain (confidence level: 100%)
Security vendors' analysis
Do you want to automate checks?
ADMINUSLabs Malicious
alphaMountain.ai Phishing
AlphaSOC Malware
Antiy-AVL Malicious
BitDefender Phishing
ESET Phishing
Forcepoint ThreatSeeker Phishing
Fortinet Malware
G-Data Phishing
Lionic Malware
Seclookup Malicious
Sophos Malicious

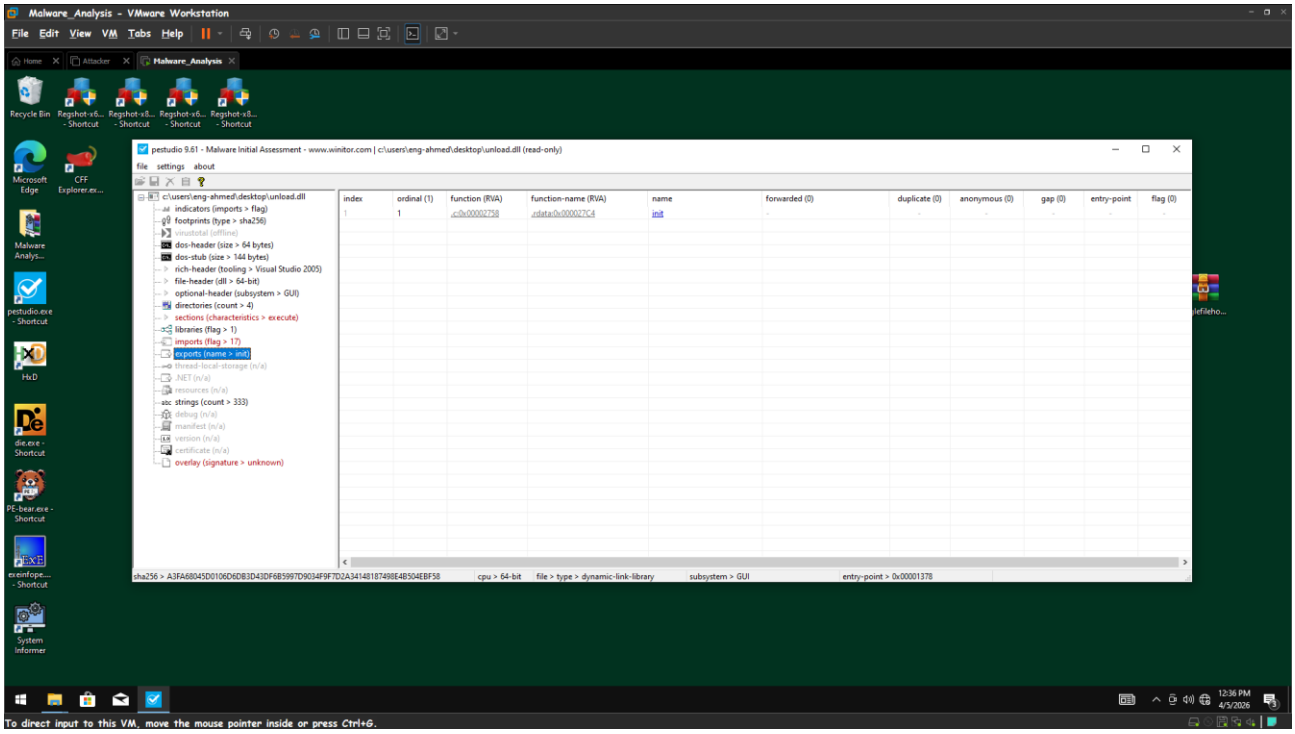

Malware_Analysis - VMware Workstation
File Edit View VM Tabs Help
Home
Attacker
Malware_Analysis
Recycle Bin
Microsoft Edge
Malware Analys...
pestudio.exe - Shortcut
HxD
die.exe - Shortcut
PE-bear.exe - Shortcut
System Informer
Select Administrator: Command Prompt
C:\Users\Eng-Ahmed\Desktop>floss --no static -- unload.dll
finding decoding function features: 100%| 37/37 [00:00<00:00, 434.93 functions/s, skipped 1 library functions (2%)]
INFO: floss.stackstrings: extracting stackstrings from 27 functions
extracting stackstrings: 100%| 27/27 [00:00<00:00, 45.38 functions/s]
INFO: floss.tightstrings: extracting tightstrings from 8 functions...
extracting tightstrings from function 0x1800028f0: 100%| 8/8 [00:02<00:00, 3.72 functions/s]
INFO: floss.string_decoder: decoding strings
INFO: floss.results: podiumstrtss.com
INFO: floss.results: podiumstrtss.com
emulating function 0x1800013ac (call 1/1): 100%| 20/20 [00:13<00:00, 1.50 functions/s]
INFO: floss: finished execution after 27.26 seconds
INFO: floss: rendering results
FLARE FLOSS RESULTS (version v3.1.1-0-g3cd3ee6)
| file path | unload.dll
| identified language | unknown
| extracted strings
| static strings | Disabled
| language strings | 0 (0 characters)
| stack strings | 0
| tight strings | 0
| decoded strings | 2
unload.dll
unload.zip
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

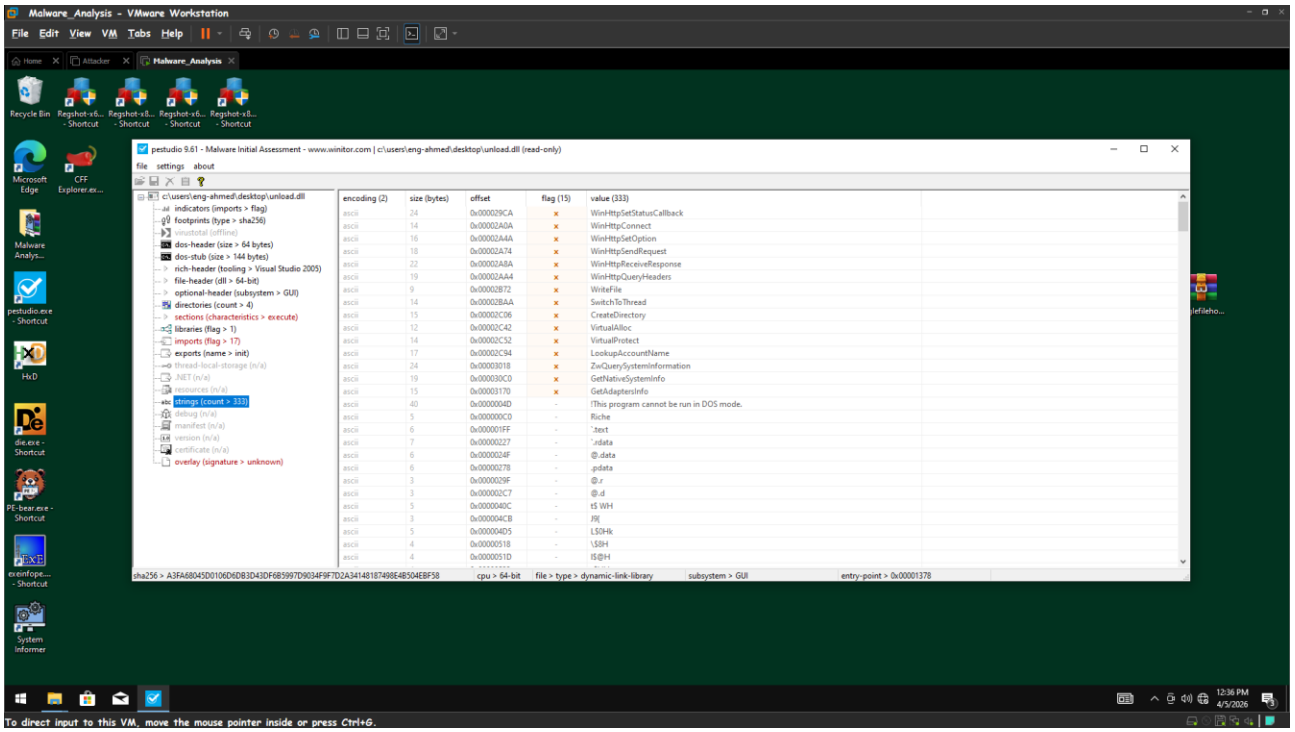

Malware_Analysis - VMware Workstation
File Edit View VM Tabs Help
pestudio 9.61 - Malware Initial Assessment - www.winitor.com | c:\users\eng-ahmed\desktop\unload.dll (read-only)
file settings about
c:\users\eng-ahmed\desktop\unload.dll
indicators (imports > flag)
footprints (type > sha256)
virustotal (offline)
dos-header (size > 64 bytes)
dos-stub (size > 144 bytes)
rich-header (tooling > Visual Studio 2005)
file-header (dll > 64-bit)
optional-header (subsystem > GUI)
directories (count > 4)
sections (characteristics > execute)
libraries (flag > 1)
imports (flag > 17)
exports (name > init)
thread-local-storage (n/a)
.NET (n/a)
resources (n/a)
strings (count > 333)
debug (n/a)
manifest (n/a)
version (n/a)
certificate (n/a)
overlay (signature > unknown)
property value
file
file > sha256 A3FA68045D0106D6DB3D43DF6B5997D9034F9F7D2A3414818749BE4B504EBF58
file > first 32 bytes (hex) 4D 5A 90 00 03 00 00 00 04 00 00 00 FF FF 00 00 B8 00 00 00 00 00 00 00 40 00 00 00 00 00 00 00
file > first 32 bytes (text) MZ.....@.....
file > info size: 16320 bytes, entropy: 4.665
file > type dynamic-link-library, 64-bit, GUI
file > version n/a
file > description n/a
entry-point > first 32 bytes (hex) 48 83 EC 38 83 FA 01 75 1F 48 83 64 24 28 00 4C 8D 05 46 15 00 00 83 64 24 20 00 45 33 C9 33 D2
entry-point > location 0x00001378 (section[.c])
file > signature Microsoft Linker 14.12 | Visual Studio 2005
stamps
stamp > compiler Wed Nov 30 18:23:22 2022 (UTC)
stamp > debug n/a
stamp > resource n/a
stamp > import n/a
stamp > export n/a
names
file > name c:\users\eng-ahmed\desktop\unload.dll
debug > file n/a
export > original-file-name loader_dll_64.dll
version n/a
manifest n/a
.NET > module > name n/a
certificate > program-name n/a
sha256 > A3FA68045D0106D6DB3D43DF6B5997D9034F9F7D2A3414818749BE4B504EBF58
cpu > 64-bit
file > type > dynamic-link-library
subsystem > GUI
entry-point > 0x00001378
singlefileho...
12:32 PM
4/5/2026
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.


Malware_Analysis - VMware Workstation
File Edit View VM Tabs Help
pestudio 9.61 - Malware Initial Assessment - www.winitor.com | c:\users\eng-ahmed\desktop\unload.dll (read-only)
file settings about
c:\users\eng-ahmed\desktop\unload.dll
indicators (imports > flag)
footprints (type > sha256)
virustotal (offline)
dos-header (size > 64 bytes)
dos-stub (size > 144 bytes)
rich-header (tooling > Visual Studio 2005)
file-header (dll > 64-bit)
optional-header (subsystem > GUI)
directories (count > 4)
sections (characteristics > execute)
libraries (flag > 1)
imports (flag > 17)
exports (name > init)
thread-local-storage (n/a)
.NET (n/a)
resources (n/a)
strings (count > 333)
debug (n/a)
manifest (n/a)
version (n/a)
certificate (n/a)
overlay (signature > unknown)
property value value value value value value value value
section section[0] section[1] section[2] section[3] section[4] section[5] section[
name .c .text .rdata .data .pdata .r .d
section > sha256 BCDFA47AE008510A78C22E... DE6AB8A8F5217E37360E609... 4086AF095BEA1BECD2B2E1E... n/a B016B81FB05B3F769623C4B... 602D01EB4B869CF93058E32... 7047B07
entropy 6.153 0.082 4.143 n/a 3.415 3.399 2.443
file > ratio (75.29%) 47.06 % 3.14 % 15.69 % n/a 3.14 % 3.14 % 3.14 %
raw-address (begin) 0x00000400 0x00002200 0x00002400 0x00000000 0x00002E00 0x00003000 0x00003
raw-address (end) 0x00002200 0x00002400 0x00002E00 0x00000000 0x00003000 0x00003200 0x00003
raw-size (12288 bytes) 0x00001E00 (7680 bytes) 0x00000200 (512 bytes) 0x00000A00 (2560 bytes) 0x00000000 (0 bytes) 0x00000200 (512 bytes) 0x00000200 (512 bytes) 0x00000
virtual-address (begin) 0x00001000 0x00003000 0x00004000 0x00005000 0x00006000 0x00007000 0x00008
virtual-address (end) 0x00002D11 0x00003006 0x000048C6 0x00005004 0x000061A4 0x000071A0 0x00008
virtual-size (10661 bytes) 0x00001D11 (7441 bytes) 0x00000006 (6 bytes) 0x000008C6 (2246 bytes) 0x00000004 (4 bytes) 0x000001A4 (420 bytes) 0x000001A0 (416 bytes) 0x00000
characteristics 0x60000020 0x60000020 0x40000040 0xC0000040 0x40000040 0x40000040 0xC000
write x x
execute x x
share
self-modifying
virtual x
items
directory > export 0x00004380
directory > import 0x000043CC
directory > exception 0x00006000
directory > import-address 0x00004000
exports > name (RVA) 0x000043B2
base-of-code 0x00001000
entry-point > location 0x00001378
sha256 > A3FA68045D0106D6DB3D43DF6B5997D9034F9F7D2A3414818749BE4B504EBF58
cpu > 64-bit
file > type > dynamic-link-library
subsystem > GUI
entry-point > 0x00001378
singlefileho...
12:32 PM
4/5/2026
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.


Malware_Analysis - VMware Workstation
File Edit View VM Tabs Help
pestudio 9.61 - Malware Initial Assessment - www.winitor.com | c:\users\eng-ahmed\desktop\unload.dll (read-only)
file settings about
c:\users\eng-ahmed\desktop\unload.dll
indicators (imports > flag)
footprints (type > sha256)
virustotal (offline)
dos-header (size > 64 bytes)
dos-stub (size > 144 bytes)
rich-header (tooling > Visual Studio 2005)
file-header (dll > 64-bit)
optional-header (subsystem > GUI)
directories (count > 4)
sections (characteristics > execute)
libraries (flag > 1)
imports (flag > 17)
exports (name > init)
thread-local-storage (n/a)
.NET (n/a)
resources (n/a)
strings (count > 333)
debug (n/a)
manifest (n/a)
version (n/a)
certificate (n/a)
overlay (signature > unknown)
imports (40) flag (17) type ordinal first-thunk (IAT) first-thunk-original (INT) library
WinHttpCloseHandle
WinHttpOpen
WinHttpConnect
WinHttpQueryHeaders
WinHttpReceiveResponse
WinHttpSendRequest
WinHttpOpenRequest
WinHttpSetOption
WinHttpQueryDataAvailable
WinHttpReadData
WinHttpSetStatusCallback
SHGetFolderPathA
memset
wsprintfW
WideCharToMultiByte
lstrlenW
VirtualProtect
VirtualAlloc
lstrcatA
lstrcpyA
GetTempPathA
CreateDirectoryA
LoadLibraryA
GetProcAddress
GetComputerNameExW
GetTickCount64
SwitchToThread
GetLastError
WINHTTP.dll
SHELL32.dll
msvcrt.dll
USER32.dll
KERNEL32.dll
sha256 > A3FA68045D0106D6DB3D43DF6B597D9034F9F7D2A341481874986E4B504EBF58
cpu > 64-bit
file > type > dynamic-link-library
subsystem > GUI
entry-point > 0x00001378
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.


Malware_Analysis - VMware Workstation
File Edit View VM Tabs Help
Detect It Easy v3.10 [Windows 10 Version 2009] (x86_64)
File name
C:\Users\Eng-Ahmed\Desktop\unload.dll
File type
PE64
File size
15.94 KiB
Base address
0000000180000000
Entry point
0000000180001378
File info
MIME
Memory map
Disasm
Hex
Strings
Signatures
VirusTotal
Visualization
Search
Hash
Entropy
Extractor
YARA
PE
Export
Import
Resources
.NET
TLS
Overlay
Manifest
Version
Sections
0007
Time date stamp
2022-11-30 20:23:22
Size of image
00009000
Scan
Automatic
Endianness
LE
Mode
64-bit
Architecture
AMD64
Type
DLL
PE64
Operation system: Windows(Vista)[AMD64, 64-bit, DLL]
Linker: Microsoft Linker(14.12.25830)
Compiler: Microsoft Visual C/C++(19.12.25830)[C]
Language: C
Tool: Visual Studio(2017, v15.5)
Overlay: Binary(Offset=0x3400,Size=0x0bc0)
Unknown: Unknown
Signatures
Flags
Database
Directory
Log
186 msec
Scan
Advanced
Demangle
Shortcuts
Options
About
Exit
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.




---

## 4. Execution Mechanism (Export Analysis)

| Field | Value |
| --- | --- |
| Exported Function | init |
| RVA | 0x00002758 |

**Execution Method**

rundll32.exe unload.dll,init

**Analysis Notes**

- Malware relies on **DLL execution via rundll32**
- Exported function acts as **entry point for payload execution**
- Common technique for **living-off-the-land execution (LOLBins)**

---

**5. Import Address Table (IAT) Analysis**

**A. Network Communication (WINHTTP.dll)**

- WinHttpOpen
- WinHttpConnect
- WinHttpOpenRequest
- WinHttpSendRequest
- WinHttpReceiveResponse
- WinHttpReadData
- WinHttpQueryHeaders
- WinHttpQueryDataAvailable

**Conclusion:**
Indicates full implementation of **HTTP-based C2 communication**

---

**B. Memory & Execution Control (KERNEL32.dll)**

- VirtualAlloc
- VirtualProtect

- LoadLibraryA
- GetProcAddress
- CreateDirectoryA

**Conclusion:**

- Enables **in-memory execution**
- Supports **dynamic API resolution**
- Suggests **file system interaction capability**

---

**6. String Deobfuscation (FLOSS Analysis)**

**Tool Used**

floss --no static -- unload.dll

**Results**

- Decoded function: 0x1800013ac
- Extracted strings: **2**
- Key Indicator:

podiumstrtss.com

**Analysis Notes**

- Heavy **string obfuscation detected**
- FLOSS confirms **runtime decoding routines**
- Extracted domain acts as **C2 indicator**

---

**7. Threat Intelligence Correlation**

**Indicator of Compromise (IOC)**

**Type Value**

Domain podiumstrtss.com

**Reputation Analysis**

- Detection Rate: **14 / 95 vendors**
- Classified as:
  - **IcedID C2 Infrastructure**
  - **Botnet Communication Endpoint**

**Attribution**

- Linked to **IcedID (BokBot) malware operations**
- Confirmed by multiple threat intelligence sources

---

### 8. Behavioral Assessment

Based on static analysis, the malware likely performs:

1. **Initial Execution**
   - Triggered via rundll32.exe
2. **C2 Communication**
   - Connects to remote domain via HTTP
3. **Payload Retrieval**
   - Downloads secondary payload
4. **In-Memory Execution**
   - Uses VirtualAlloc + VirtualProtect
5. **Evasion Techniques**
   - Obfuscation
   - Dynamic API resolution
   - Non-standard PE structure

---

### 9. Conclusion

The analyzed DLL is a **malicious loader** strongly associated with the **IcedID malware family**.

It demonstrates:

- **Advanced evasion techniques**
- **Network-based payload delivery**
- **Fileless execution behavior**

This confirms its role as a **Stage-1 loader** designed to:

- Establish persistence
- Retrieve additional malware components
- Execute payloads stealthily in memory

---

**10. Dynamic Analysis Plan**

**Monitoring Strategy**

**1. Sysmon Configuration**

- Event ID 1 → Process Creation (rundll32.exe)
- Event ID 3 → Network Connections
- Event ID 11 → File Creation

**2. Process Monitoring**

- Tool: Process Hacker
- الهدف:
    - Identify **RWX memory regions**
    - Dump decrypted payload

**3. Debugging Strategy**

- Tool: IDA Pro Debugger
- Breakpoints:
    - WinHttpSendRequest → Capture beacon
    - VirtualProtect → Intercept unpacked payload

---

## 11. Indicators of Compromise (Summary)

| Type | Indicator |
| --- | --- |
| File Hash | A3FA68045D0106D6DB3D43DF6B5997D9034F9F7D2A34148187498E4B504EBF58 |
| Domain | podiumstrtss.com |
| Execution | rundll32.exe unload.dll,init |

---

### Final Assessment

This sample represents a **high-confidence malicious artifact** associated with **financial malware operations (IcedID)** and should be:

- Blocked at **endpoint and network level**
- Added to **SIEM detection rules**
- Used for **threat hunting across environment**

---